

A Machine Learning based approach for Dynamic Hand Gesture Recognition in Human-Robot Interaction

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Abstract— Today's modern world is equipped with many smart robotic gadgets and devices to make our day-to-day jobs easier. These robotic devices need to capture human orders using different kinds of audio-visual inputs, hand gestures being one of them. Therefore, hand gesture recognition is an important step in human-robot interaction. The response of the robot directly depends on the time taken in classification of the performed gesture by the human. However, processing all the redundant frames in the visual input increases the response time of the device. In this paper, we propose a machine learning based approach to dynamic hand gesture recognition, by extracting only unique key-frames from the input visuals. To obtain them, firstly the hand skeleton landmarks are focused in order to obtain the important spatial features. The similarity in these landmark features is analyzed and used to determine the unique key-frames from each input video. The unique key-frames so decided by similarity detection are further input to SVM machine learning model to perform hand gesture classification, achieving a classification accuracy of 79.30% at 20 key-frames within a training time of 1241 sec, which is comparable to the one using all frames, taking a training time of 6455 sec. The obtained results have been further compared with regular feature extraction method using PCA, showing improved results of proposed method over PCA.

Keywords— human-robot-interaction, dynamic-hand-gesture-recognition, machine-learning, key-frame-extraction

I. INTRODUCTION

Smart Robotic devices use different kinds of A/V input from a user to give instructions to the robotic device, like speech input [10], facial input [11], environmental constraints [9], hand gesture being an important one of them. The general steps involved in Human-Robot interaction by dynamic hand gestures are shown in figure 1.1. The dynamic hand gestures captured in the form of video can be used to capture both spatial and temporal features of the input, rather than only using static images [18]. The captured video gesture has to be first classified into the corresponding gesture class so that the robotic device can perform the corresponding action. Thus, a machine learning or deep learning model can be trained to perform dynamic hand gesture recognition. However, the video input contains a lot of redundant information because there are many similar frames in an HD video [8]; in order to obtain the essence of a hand gesture, only the unique frames, called key-frames can be used to identify the hand gesture. Selection of key-frames further reduces the training load on the otherwise complex machine learning and deep learning systems [12]. Moreover, the quality and uniqueness of the so extracted key-frames further effects the classification accuracy of the model and thus the further response of a device. Various Key-Frame extraction methods have been proposed by researchers [6], [7] to obtain the best possible key-frames from an input video.

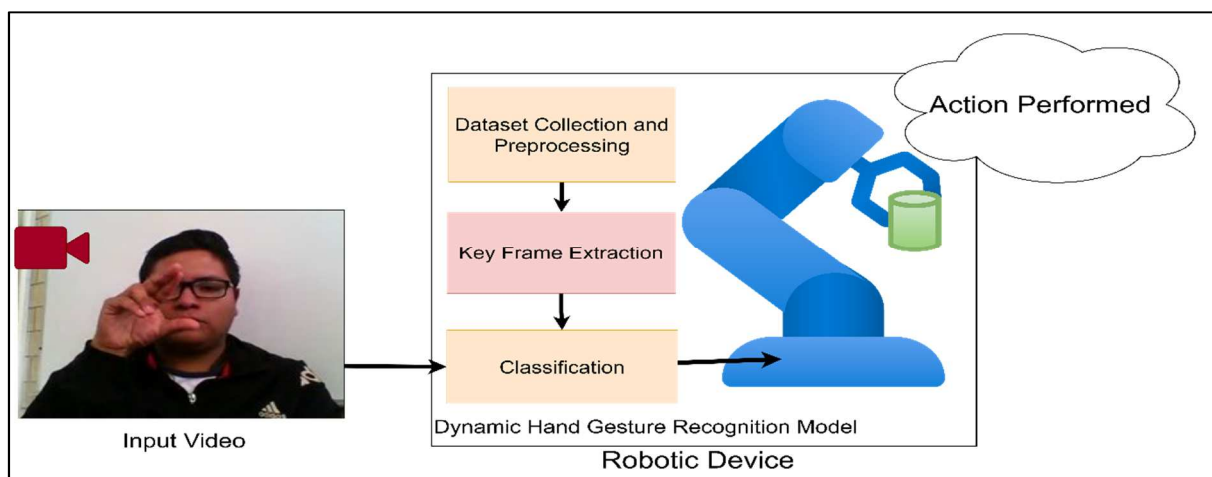


Figure 1.1: Basic steps in Human-Robot Interaction

Thus, the central problem of this paper is to achieve accurate classification results from the input video samples using the minimal number of unique key-frames from the input video, to be effectively used for obtaining the desired action from a robotic

device. For this, a frame similarity-based K-Means approach has been proposed to first identify the unique key-frames, and then further classified using SVM classifier.

The remaining of this article has been organized as follows: Section II provides a brief literature review of the recent works in the domain under consideration. Section III explains the detailed methodology of the proposed model. Section IV gives the detailed results and discussion, followed by Conclusion and Future Scope in Section V.

II. LITERATURE REVIEW

The previous decade has seen a surge in the research related to dynamic hand gesture recognition. Various researchers have been using dynamic hand gesture recognition as an important phase in different applications such as sign language interpretation, human computer interaction, human-machine and robot interaction, AR/VR devices, smart appliance and drone control etc. Authors in [1] have developed dynamic hand gesture benchmark dataset named IPN Hand containing dynamic hand gesture videos from various subjects, and also proposed CNN classification mechanism for dynamic hand gesture recognition from the prepared dataset. Authors in [5] have developed a human-robot interaction model using dynamic hand gesture video input. Neural network classification has been performed using different distance and time features within hand landmarks. [13] proposes a dataset of dynamic hand gesture recognition named DATE dataset for use in daily conversation and also builds deep-learning based benchmark models for its classification. Authors in [2] have proposed a 3DCNN based dynamic hand gesture recognition model on the EgoGesture dataset. An improved interframe difference method has been proposed for performing key-frame extraction, achieving an overall accuracy of 72.4%. [3] proposes a vision based dynamic hand gestures recognition method using short-term and long-term neural networks. The NVIDIA dataset has been used to prove the effectiveness of the proposed methodology. Random frame selection method has been used on groups of frames. Authors in [4] have implemented alphabet and numerals recognition from hand sign videos. Gesture trajectory features have been extracted and classification has been performed using six different classifiers. An accuracy of 86.97% has been achieved. Article in [14] has proposed an ANN and SVM based model for recognition of hand gestures from static images. MS Kinect and depth sensors have been used to perform image extraction. Alphabet images have been classified using extracted skeleton points. Authors in [15] have proposed a gesture based human robot interaction model by computer vision methods. Two step modules have been implemented to utilize local and global features in hand gestures. [16] proposes a dynamic hand gesture recognition model from human robot interaction from long distances using GCN and vision transformer. Testing

has been performed on diverse datasets achieving average accuracy of 96%.

III. METHODOLOGY

The detailed methodology followed in the process of dynamic hand gesture recognition has been shown in figure 3.1. The steps involved in the proposed methodology have been elaborated as follows:

A. Preprocessing

An initially captured video from an RGB camera is actually a sequence of image frames at a particular frame rate. The first and foremost step in the proposed methodology is to break down the input video into their constituent frames at the original frame rate so that the next steps in the model can operate on them. Here, the frames from each video are separated out, yet they are sorted in the same time sequence as the original video, and stored in a common directory structure so that all the frames from a single video can be viewed as a single sample during the further steps. Each frame of the videos is resized to 64*64 standard dimensions across the dataset.

B. Hand Landmark Detection

The pre detection of different hand skeleton points in the input frames can significantly help in analysing the hand motion by the machine learning model. In the next step, the hand skeleton points in the image frames are detected using the media-pipe library. The mpyhnds function in the media-pipe library returns 21 landmark points for each hand image in the form of their x, y and z coordinates as shown in the figure 3.2. The corresponding x-y-z coordinates of the detected landmark locations are used for further processing.

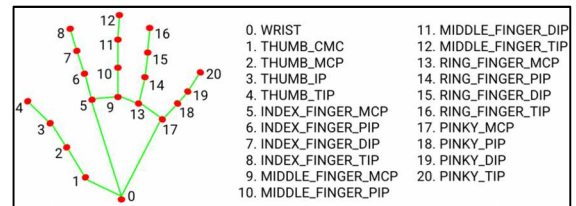


Figure 3.2: Hand landmarks returned by media-pipe [17]

C. Frame Filtering

As the hand landmark points are detected in the input frames, some of the frames may not contain any hand image. Such frames can be eliminated from the further process because they will not be relevant in the hand gesture recognition process. Such frames will not have any coordinate values for the hand landmark points and are indicated red in the figure 3.1. Thus, we exclude those frames for the future steps and obtain a reduced feature vector as shown in figure 3.1.



Figure: 3.1 Proposed Model

D. Similarity Detection

In the next phase, the similarity among the input frames is detected by calculating the pair wise Euclidean distance between the hand landmark coordinates in different frames, marked using similar color codes in figure 3.1. Given two frames P and Q with landmark coordinates ranging from $p_1 \dots p_l, q_1 \dots q_l$, the pair wise Euclidean distance may be calculated as (1):

$$D_{pq} = \sqrt{\sum_{i=1}^l (p_i - q_i)^2} \quad (1)$$

If the pair-wise Euclidean distance within two frames is below a threshold value, they remain in the same cluster, else they are allocated a separate cluster in the K-Means mechanism. This leads to allocation of the frames with similar hand positions in the same cluster.

E. Unique Frame Selection

After the clustering of similar frames in separate clusters, the frame having minimum pair wise Euclidean distance from each cluster centroid is

selected as a key-frame K from that cluster given as (2).

$$K = \operatorname{argmin}(\sum_{p=1}^N \sum_{q=1}^{n_i} (D_{pq})^2) \quad (2)$$

Eventually, selecting only one key-frame from each cluster leads to unique frame selection from the input frames in a video. Thus, the selected number of key-frames equals the value of K used in K-Means clustering.

F. Feature Vector Flattening

Since our single input sample corresponds to a complete video, in this phase the landmark features of the above selected key-frames are flattened into a single row along with the corresponding gesture class to be compatible with further machine learning steps. The number of features ' f ' eventually contained in the flattened feature vector with K key-frames, with l_k hand landmarks each, each having their own x, y and z coordinate values may be computed as (3).

$$f = \sum_{k=1}^K (3 * l_k) \quad (3)$$

G. Train-Test Split

The final flattened feature vector is now split into train and test data in 75:25 train-test ratios to be used for further training and testing in SVM machine learning model.

H. SVM Classification

Finally, the train data is used for training the SVM machine learning model for classification of dynamic hand gestures. Eventually, testing is performed using the test dataset, and the corresponding gestures are interpreted to perform various functions in a robotic device.

IV. EXPERIMENTATION AND RESULTS

Python version 3.11.4 with CUDA V12.1.105, Keras 2.10.0, and TensorFlow 2.10.1 libraries has been used for all experiments on Anaconda Spyder 5.4.1. The training, validation, and testing of the designed model were performed on an x-64-based 12th Gen Intel(R) Core(TM) i7-12700H 2.30 GHz processor with 16 GB system RAM and 64-bit Windows 11 operating system. The system contains a single NVIDIA GeForce RTX 3050 GPU with 4GB of dedicated memory.

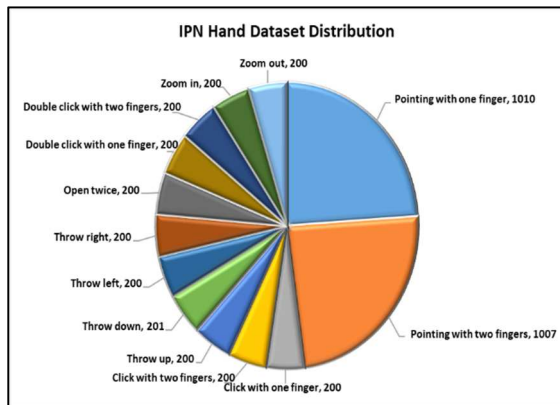


Figure 4.1: IPN Hand class wise distribution



Figure 4.2: Sample Key-frames in gesture classes

A. Dataset Used

The dataset used for the training and testing of the proposed model has been obtained from publicly available IPN hand dataset. It contains 13 classes of hand gestures comprising 2 static and 11 dynamic hand gestures that can be used in human-machine interaction. The class wise distribution of

samples in the IPN hand dataset has been shown in Figure 4.1. Samples of the key-frames representing some gesture classes have been shown in figure 4.2.

B. Results and Discussion

The results of the proposed method have been obtained by varying the value of K from 5 to 20 keeping a standard interval. The classification accuracy so obtained has been compared with that obtained using all the input frames. It has been observed that the classification accuracy obtained by using only 20 key-frames surpasses the classification accuracy when using all the input frames. The best achieved accuracy has been notified at K=20, with 20 selected key-frames from each video, as 79.30%. It may be noted that the classification accuracy no longer improves when the number of key-frames is increased beyond 20. A comparison of the accuracy achieved with different values of K and that without using any key-frame selection has been shown in figure 4.3.

Further, we also compare the training time taken in the different scenarios. It has been analyzed that as compared to the all the input frames taking a training time of 6455 sec, the accuracy using 20 key-frames surpasses it only in a training time of 1241 sec as shown in figure 4.4.

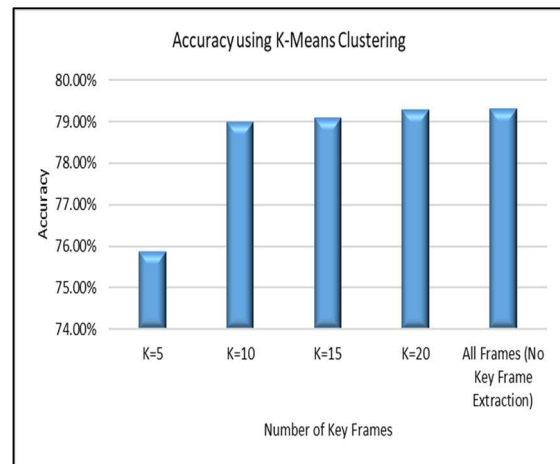


Figure 4.3: Accuracy of proposed model

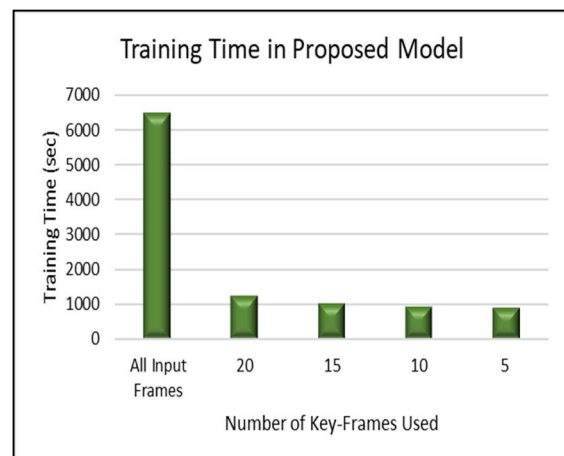


Figure 4.4: Training time in proposed model

Figure 4.5 depicts the confusion matrix so obtained by testing the trained SVM model using 20 key-frames for classification.

True Class	B0A	239	3	1		1		1		2	3		1	1
	B0B	12	229	2	1			2	4			1		
	G01			26	1						23			
	G02	1	3	2	24				1		2	13	2	2
	G03			1	1	37		3		4		1	1	2
	G04	1	4	3	1	6	31		1	1		1		2
	G05			1		5	2	41	1	1				
	G06		2	1			1		40	4		1		1
	G07		2	1	1	1	1		2	37		2	2	1
	G08	3		12						1	31		1	1
	G09		3			15				1	3	20	5	3
	G10		3	1		4			1	4		2	33	2
	G11	1	2	1		3	1				3	5	3	29
	B0A	B0B	G01	G02	G03	G04	G05	G06	G07	G08	G09	G10	G11	

Figure 4.5 Confusion Matrix

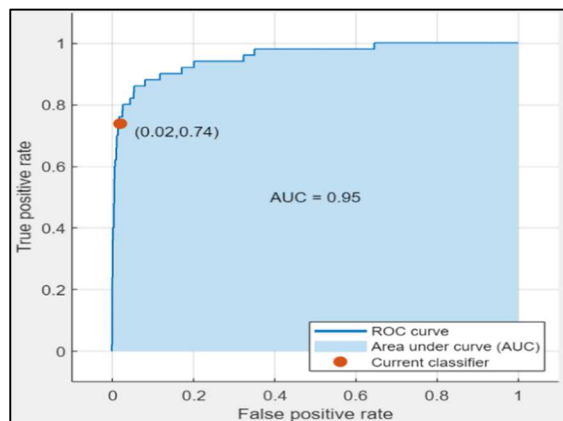


Figure 4.6: Area under ROC curve

Also, figure 4.6 depicts the corresponding ROC curve with a 95% area under ROC curve, proving a high true positive rate against the false positive rate in classification.

Further, in order to verify the superiority of the proposed model, it has been compared to the traditional feature extraction method using PCA. The variance of the PCA has been kept from within 95% to 80% and the achieved accuracy has been analyzed. The PCA with SVM classification achieved its best accuracy of only 64.10%, by keeping a variance of 95% within the extracted features. Although, it has been noted that only 38 features were extracted using PCA as that compared to 1260 features selected by the proposed system, however the achieved accuracy gap in the proposed method verifies its superiority over the PCA approach.

The results so obtained by different variance values in PCA have been depicted in figure 4.7, further figure 4.8 compares the best accuracy so obtained by the proposed method against the PCA.

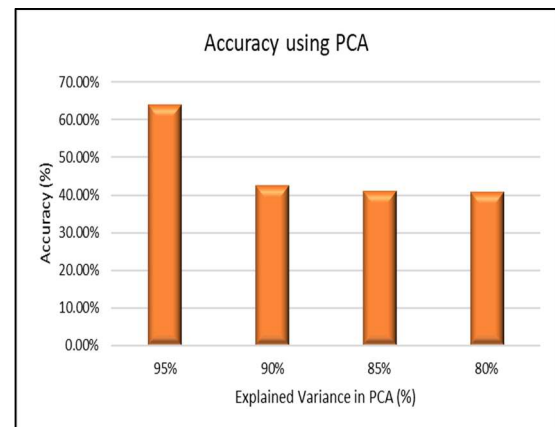


Figure 4.7: Accuracy using PCA

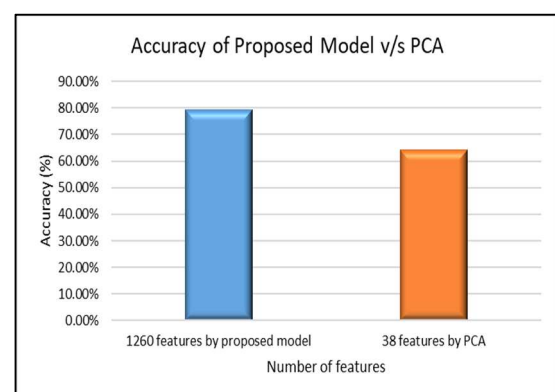


Figure 4.8: Accuracy of proposed model v/s PCA

V. CONCLUSION AND FUTURE SCOPE

This article proposes an SVM machine learning based dynamic hand gesture classification approach as an application to human-robot interaction. A frame similarity-based clustering approach has been proposed to firstly identify the unique frames from an input video, leading to redundancy removal. Further, only the so extracted key-frames have been used to perform SVM based machine learning classification of hand gestures. The IPN hand benchmark dataset of dynamic hand gestures has been used for testing the proposed approach, achieving a notable classification accuracy of 79.30% using only 20 key frames per video, as compared to using all the input frames, in a quite less training time of 1260 sec. The results so obtained have also been compared to tradition feature extraction method using PCA, showing superiority of the proposed model.

The future scope of the proposed approach lies in further improving the key-frame extraction approach to incorporate dynamic behavior of video data. Moreover, testing the classification performance using the state of art deep learning approaches by using the key-frames extracted in this approach is also a dimension worth exploring in future.

VI. REFERENCES

- [1] Benitez-Garcia, Gibran, et al. "IPN hand: A video dataset and benchmark for real-time continuous hand gesture recognition." 2020 25th international conference on pattern recognition (ICPR). IEEE, 2021.
- [2] Liu, Yuting, et al. "Dynamic gesture recognition algorithm based on 3D convolutional neural network." Computational Intelligence and Neuroscience 2021 (2021).
- [3] Zhang, Wenjin, Jiacun Wang, and Fangping Lan. "Dynamic hand gesture recognition based on short-term sampling neural networks." IEEE/CAA Journal of Automatica Sinica 8.1 (2020): 110-120.
- [4] Saboo, Shweta, Joyeeta Singha, and Rabul Hussain Laskar. "Dynamic hand gesture recognition using combination of two-level tracker and trajectory-guided features." Multimedia Systems 28.1 (2022): 183-194.
- [5] Peral, Marc, Alberto Sanfeliu, and Anaís Garrell. "Efficient hand gesture recognition for human-robot interaction." IEEE Robotics and Automation Letters 7.4 (2022): 10272-10279.
- [6] Aiswarya, M. S., and R. Arockia Xavier Annie. "Key-frame Extraction Algorithm for Continuous Sign-Language Videos Using Angular Displacement and Sequence Check Metrics." International Journal of Intelligent Systems 2024 (2024).
- [7] Ansar, Hira, et al. "Dynamic Hand Gesture Recognition for Smart Lifecare Routines via K-Ary Tree Hashing Classifier." Applied Sciences 12.13 (2022): 6481.
- [8] Sunuwar, Jhuma, and Samarjeet Borah. "A comparative analysis on major key-frame extraction techniques." Multimedia Tools and Applications (2024): 1-46.
- [9] Hoebert, Timon, et al. "Knowledge-driven framework for industrial robotic systems." Journal of Intelligent Manufacturing 34.2 (2023): 771-788.
- [10] Teeda, Vineeth, K. Sujatha, and Rakesh Mutukuru. "Robot voice a voice controlled robot using arduino." arXiv preprint arXiv:2402.03803 (2024).
- [11] Grover, Richa, and Sandhya Bansal. "CNN-based human emotion recognition from facial images in controlled-uncontrolled environment." 2023 Second International Conference on Informatics (ICI). IEEE, 2023.
- [12] Tyagi, Akansha, and Dr Sandhya Bansal. "Hybrid FAST-SIFT-CNN (HFSC) approach for vision-based Indian sign language recognition." International Journal Of Computing and Digital System (2021).
- [13] Dang, Tuan Linh, et al. "DATE: a video dataset and benchmark for dynamic hand gesture recognition." Neural Computing and Applications (2024): 1-15.
- [14] Bamwenda, Julius, and Mehmet Sirac Özerdem. "Recognition of static hand gesture with using ANN and SVM." Dicle Univ. J. Eng 10 (2019): 561-568.
- [15] Zhou, Guoyu, Zhenchao Cui, and Jing Qi. "FGDSNet: A Lightweight Hand Gesture Recognition Network for Human Robot Interaction." IEEE Robotics and Automation Letters (2024).
- [16] Bamani, Eran, et al. "Ultra-Range Gesture Recognition using a web-camera in Human-Robot Interaction." Engineering Applications of Artificial Intelligence 132 (2024): 108443.
- [17] Tyagi, A.: Sandhya B.: Sign language recognition using hand landmark analysis for vision-based system (HMASL). Emergent converging technologies and biomedical systems: Select proceedings of ETBS 2021. Singapore: Springer Singapore, 431-445 (2022)
- [18] Kaur, Arpneek, and Sandhya Bansal. "Deep learning for dynamic hand gesture recognition: applications, challenges and future scope." 2022 5th International Conference on Multimedia, Signal Processing and Communication Technologies (IMPACT). IEEE, 2022.